

scReg-Eval: a fixed-panel audit of regulatory alignment in single-cell RNA foundation-model gene graphs

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Abstract

Claims that single-cell RNA foundation models encode gene regulation are difficult to audit because many reference networks share co-expression structure with the models' training data. We introduce **scReg-Eval**, a fixed-panel protocol that compares foundation-model gene graphs with a sequence- and accessibility-derived regulatory-potential proxy while controlling co-expression, target peak count, gene length, GC content, RNA detection rate, transcription-factor out-degree, and target in-degree. The preregistered panel contains 1,200 genes, including 446 in-range transcription factors. We evaluate thirteen pooled model/readout combinations across brain and paired PBMC multiome data using two plus-one Monte Carlo randomization tests ($N = 999$ each): a gene-label permutation and a within-TF-row, degree-preserving target shuffle. Benjamini–Hochberg correction is applied within eight prespecified tissue-by-confound-specification-by-null families. Under the primary full confound specification, seven of thirteen rows are supported by both nulls: six small positive alignments (Geneformer embedding in brain and PBMC, UCE encoder in brain and PBMC, scGPT encoder in brain and PBMC; ρ from 0.0043 to 0.0124) and one negative alignment (Geneformer attention in brain, $\rho = -0.0036$) from the same model whose embedding readout is positive in the same tissue. The co-expression baseline, tested on the same edge set, is itself supported in both tissues, and the PBMC Geneformer-embedding alignment ($\rho = 0.0043$) is smaller than its own baseline ($\rho = 0.0070$). Under the non-degree specification, no row is supported by both nulls, and three of the six positives change sign, showing that the alignments depend on conditioning on proxy-derived degree covariates. A supervised TF-disjoint probe gives mixed family-level results: UCE versus co-expression is supported by paired sign-flip testing ($q = 0.0325$), as is the random-init floor ($q = 0.0300$), whereas Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) are not. The probe therefore must not be summarized as showing that all models are indistinguishable from co-expression. The audit is consistent with weak, specification-dependent alignment concentrated in embedding-geometry readouts, and does not support a global regulatory-capability claim.

1 Introduction

Single-cell RNA foundation models (scFMs) are increasingly used as frozen representations for network inference and perturbation analysis [4, 13]. A high score against a curated or transcriptome-derived regulatory network, however, need not imply that a model encodes regulation beyond co-expression. Both the reference and the model can inherit gene abundance, detectability, hubness, and co-variation.

We address this problem as an audit-design question. scReg-Eval freezes a TF–target panel, constructs an accessibility-and-motif regulatory-potential proxy without expression covariation, ap-

plies the same edge mask and confounds to every model, and evaluates the observed alignment against two structurally different randomization nulls. This design does not make the proxy causal ground truth. It asks a narrower, reproducible question: on the fixed panel, does a model gene graph align with the proxy after controlling measured RNA and graph structure, and is that alignment unusual under both label and target-assignment randomizations? Negative benchmarks of foundation-model interpretability under careful controls have precedent [2]; our aim is the audit protocol and its reporting discipline rather than a verdict by itself.

Our contributions are:

1. a hash-pinned 446-TF by 1,200-gene audit panel and a sequence/accessibility regulatory-potential proxy that is observed to reproduce across brain, blood, and fibroblast datasets;
2. a full-confound partial-Spearman statistic that controls co-expression and six structural covariates, with a non-degree specification reported separately as sensitivity analysis;
3. two finite-panel Monte Carlo tests with explicit null semantics, deterministic seed streams, plus-one correction, and eight prespecified FDR families;
4. a multi-readout audit of Geneformer, scGPT, scFoundation, and UCE [4, 6, 12, 13] showing small, readout-specific primary alignments rather than a uniform positive or negative result; and
5. descriptive per-cell-type, cross-tissue, and axis-aligned injection diagnostics whose scope is kept separate from the pooled inferential families.

scReg-Eval reporting checklist. Freeze and hash the TF–target panel; label the reference as a regulatory-potential proxy rather than causal truth; control co-expression and structural covariates; state exactly what each randomization changes; use plus-one Monte Carlo p -values; define FDR families before reading results; and keep exploratory cell-type and sensitivity analyses distinct from the primary test.

2 Methods

2.1 Regulatory-potential proxy and fixed panel

The panel contains 1,200 genes, including 446 unique in-range transcription factors, frozen by a SHA-256 manifest. Accessible peaks from brain snATAC-seq, paired PBMC multiome, and fibroblast ATAC are linked to genes by a ± 2 kb promoter-plus-gene-body rule. JASPAR2024 CORE vertebrate motifs are scanned with MOODS [8, 11]; motif evidence in an accessible peak contributes a directed TF–target regulatory-potential weight. The graph uses no expression covariation, but accessibility, motif occurrence, and proximity do not establish direct regulation or causality. We therefore call it a proxy throughout.

Brain analyses exclude edges incident to a prespecified neural marker set; the same mask is applied to every brain readout. PBMC edges are unmasked under the prespecified tissue policy. Before analysis, the driver verifies the manifest, TF identity and order across all caches, matrix dimensions and finiteness, and four pinned legacy-file hashes.

2.2 Foundation-model and baseline graphs

Gene graphs are read from cached, model-validated outputs over the same manifest. Available pooled readouts are Geneformer embedding, attention, and in-silico knockout; scFoundation encoder; UCE encoder; and scGPT encoder; plus a random-init Geneformer floor. Coverage differs by tissue because only precomputed, validated graphs are included: brain has eight FM/readout rows and PBMC has five. Co-expression is reported as a baseline on the same edge set and nulls, but is not included as an FM row or in FM multiplicity families. CellPLM is outside scope because its architecture does not expose contextual per-gene states [14].

2.3 Observed statistic and confound specifications

For each retained TF–target pair, all graph weights are rank transformed. The primary statistic is the Pearson correlation between residualized ranks, equivalent to partial Spearman correlation. The *full* design contains an intercept, ranked co-expression, and standardized target peak count, gene length, RNA detection rate, GC content, TF out-degree, and target in-degree. The two degree covariates are computed from the proxy graph itself, so the full specification conditions on functions of the outcome; this is a validity hazard, not a bookkeeping detail, and it is why the *non-degree* design, which omits both degree columns, is reported co-primary rather than as a footnote. Observed and randomized statistics always use matching column sets.

2.4 Finite-panel randomization inference

The TF panel is treated as fixed; no TF-superpopulation bootstrap [5] or confidence interval is used. For the gene-label null, one permutation relabels both endpoints of the proxy graph, in the spirit of the Mantel matrix correspondence test [9]. The FM graph, co-expression, and non-degree gene covariates remain fixed. In the full specification, proxy-derived TF out-degree and target in-degree are recomputed after relabeling.

The second null independently shuffles edge weights within each TF row across non-self targets. This preserves TF out-degree and the row’s weight multiset while changing target assignment; target in-degree is recomputed only for the full specification. One proxy randomization per replicate is shared across model rows, with each model scored against the same replicate. Explicit-index tests verify numerical equivalence to separate least-squares calculations, including rank-deficient designs.

Both tests use two-sided plus-one correction,

$$p_{MC} = \frac{1 + \sum_{b=1}^{999} \mathbf{1}\{|T_b| \geq |T_{\text{obs}}|\}}{1000},$$

with resolution 0.001. Benjamini–Hochberg correction [3] is performed within each of eight families: tissue (brain/PBMC) $\times$ confound specification (full/non-degree) $\times$ null (gene-label/row-shuffle). The full specification is the preregistered primary; the non-degree specification is reported alongside it as co-primary because the full specification conditions on proxy-derived degree covariates.

2.5 TF-disjoint supervised probe

The pooled audit is an unsupervised raw-readout test. As a supervised complement, we train a ridge probe to predict each TF’s proxy-target profile from the FM graph itself. Samples are

ordered (TF, gene) pairs; features for each family are the edge weight $S_{tf,g}$, its reverse $S_{g,tf}$, and the within-TF rank of the edge, so nothing in the design is indexed by TF identity and a probe trained on 311 held-in TFs transfers unchanged to 135 disjoint test TFs. Regularization is chosen by cross-validation on the training TFs. Performance is the per-test-TF Spearman correlation between predicted and observed proxy profiles, averaged over test TFs, after residualizing both sides against target peak count, gene length, GC content, and detection rate. The null permutes gene labels within each test TF; paired FM-versus-co-expression contrasts use sign-flipped per-TF differences. Gene-level degree features are withheld from the primary probe arm because they let the probe reach the proxy through gene-level confounding rather than through the graph’s edges.

2.6 Descriptive and pipeline-sensitivity analyses

Per-cell-type rows report observed partial correlations, cell counts, ranges, medians, and sign counts. They have no randomization p -values or FDR adjustment. Cross-tissue proxy reproducibility is also observed-only. Finally, for each tissue we evaluate the following injected fractions:

$$\alpha \in \{0, .02, .05, .08, .12, .18, .25, .35, .5, .75, 1\}.$$

For each value, we combine an independently generated noise vector with an α fraction of the proxy residual and repeat the observed statistic 30 times. This axis-aligned diagnostic checks implementation response; it is not a power analysis and does not define an MDE, confidence interval, or exclusion bound.

3 Results

3.1 The proxy is reproducible, but remains a proxy

The accessibility/motif construct shows moderate observed agreement across tissue consensus graphs: brain–PBMC $\rho = 0.5249$, brain–fibroblast $\rho = 0.5245$, and PBMC–fibroblast $\rho = 0.4623$ (Fig. 1; Table 1). These are descriptive fixed-panel correlations without new randomization inference. They establish reproducibility of the construct, not a causal ceiling or proof of TF–target truth.

3.2 Primary full-confound audit: six weak positives, one negative, and a baseline that matches them

Table 2 and Fig. 2 report all thirteen pooled FM/readout rows plus the co-expression baseline. Six small positive effects are supported by both randomization tests after within-family BH correction. In brain, Geneformer embedding has $\rho = 0.00442$ ($q_M = 0.006$, $q_D = 0.003$), UCE encoder has $\rho = 0.00461$ ($q_M = 0.004$, $q_D = 0.003$), and scGPT encoder has $\rho = 0.01241$ ($q_M = 0.004$, $q_D = 0.003$). In paired PBMC, Geneformer embedding has $\rho = 0.00425$ ($q_M = 0.002$, $q_D = 0.003$), scGPT encoder has $\rho = 0.01098$ ($q_M = 0.002$, $q_D = 0.003$), and UCE encoder has $\rho = 0.00593$ ($q_M = 0.002$, $q_D = 0.003$).

One further row is supported by both nulls with the opposite sign: brain Geneformer attention, $\rho = -0.00362$ ($q_M = 0.005$, $q_D = 0.016$), from the same model whose embedding readout is positive in the same tissue. A sign-inconsistent pair of readouts within one model is not what a regulatory-capability account predicts.

The remaining rows are not supported: brain scFoundation ($\rho = 0.00062$), Geneformer knockout (-0.00093) and its position-control readout (-0.00189), the random-init Geneformer floor (0.00187), and PBMC Geneformer attention (-0.00018) and scFoundation (0.00112).

The co-expression baseline, run through the same nulls on the same edge set (it cannot enter the FM families because partialling co-expression out of itself is degenerate), is itself supported in both tissues: brain $\rho = 0.00276$ ($p_M = 0.019$, $p_D = 0.014$) and PBMC $\rho = 0.00697$ ($p_M = 0.001$, $p_D = 0.001$). Two observations follow. First, the baseline reaches the same order of alignment as the supported FM rows. Second, the PBMC Geneformer-embedding positive ($\rho = 0.00425$) is smaller than its own baseline ($\rho = 0.00697$): on that row the model aligns with the proxy less than the confound it is adjusted against.

3.3 The alignments depend on conditioning on proxy-derived degree covariates

Removing the two degree covariates changes both effect direction and randomization behavior (Fig. 3). No non-degree row is supported by both nulls, and three of the six full-spec positives (brain Geneformer embedding, brain scGPT, PBMC Geneformer embedding) are negative before degree conditioning. Because the degree covariates are computed from the proxy graph itself, the full specification conditions on functions of the outcome; the positives therefore appear in exactly the specification that carries this hazard. We report the non-degree specification as co-primary and treat the divergence as a substantive finding, not a footnote: the row-shuffle null alone supports several non-degree rows while the gene-label null supports none, so degree adjustment and null choice define different finite-panel questions.

3.4 Per-cell-type patterns are descriptive

The full-confound cell-type analysis contains 15 brain and 14 PBMC FM rows per specification (Fig. 5). Brain full-confound estimates range from -0.00396 to 0.00445 with median 0.00258 ; PBMC estimates range from 0.00222 to 0.00682 with median 0.00413 . These rows are exploratory robustness summaries without p -values or BH correction. They neither promote cell-type discoveries nor provide population uncertainty. We also note that the proxy itself is near cell-type-invariant on this panel (pairwise consensus $\rho \approx 0.99$), so per-cell-type rows score readouts against a reference with essentially no cell-type contrast.

3.5 The implementation responds monotonically to aligned injection, and the observed effects sit at the bottom of that ladder

The axis-aligned diagnostic rises monotonically from approximately zero at $\alpha = 0$ to mean partial correlations of 0.905 in brain and 0.913 in PBMC at $\alpha = 1$ (Fig. 4). Subdividing the ladder at $\alpha \in \{0.002, 0.005, 0.01\}$ places the observed effects on the curve: the pipeline recovers mean $\rho \approx 0.002$, 0.004 – 0.005 , and ≈ 0.010 at those fractions, so the supported rows are equivalent to injections of roughly 0.004 – 0.013 of the proxy’s own residual structure. This diagnostic checks implementation response; it is not a power analysis and does not define an MDE, confidence interval, or exclusion bound.

3.6 A supervised TF-disjoint probe gives mixed family-level results

A linear probe trained to predict each TF’s proxy targets from graph features on 311 held-in TFs and tested on 135 disjoint TFs (Fig. 6) gives mixed family-level results after confound adjust-

ment. Paired sign-flip testing supports the UCE-versus-co-expression contrast ($q = 0.0325$) and the random-init floor contrast ($q = 0.0300$), while the Geneformer embedding ($q = 0.5125$), Geneformer attention ($q = 0.65$), and scGPT ($q = 0.255$) contrasts are not supported. The adjusted test correlations span -0.019 to $+0.002$. The probe’s confound set differs from the audit’s—it contains neither co-expression nor the proxy-derived degree covariates—so it answers a different question (supervised recoverability, not raw-readout alignment) and neither confirms nor refutes the pooled rows. In particular, the random-init result prevents interpreting the UCE contrast as model-specific regulatory recovery, while the UCE result prevents a blanket claim that every FM is indistinguishable from co-expression.

4 Discussion

The audit finds six small positive fixed-panel alignments and one negative one, all supported by both randomization nulls under the full confound specification. Three features of that result restrain its interpretation. First, the co-expression baseline, tested on the same edge set, is itself supported in both tissues, and one of the six positives (PBM Geneformer embedding) is smaller than its own baseline; the models do not separate from the confound they are adjusted against. Second, the two supported readouts of the same model in the same tissue (Geneformer embedding $+0.0044$, Geneformer attention -0.0036 , both brain) have opposite signs, which is not the signature of a model-level regulatory capability. Third, three of the six positives are negative before conditioning on the proxy’s own degree covariates, so they appear only in the specification that conditions on functions of the outcome. A supervised TF-disjoint probe, answering the distinct question of supervised recoverability, finds mixed contrasts: UCE and the random-init floor are supported against co-expression ($q = 0.0325$ and $q = 0.0300$), while Geneformer embedding, Geneformer attention, and scGPT are not ($q = 0.5125$, $q = 0.65$, and $q = 0.255$). The random-init result cautions against reading the UCE contrast as model-specific regulatory recovery.

The most defensible reading is therefore not “four models encode weak regulation.” It is that small, readout-specific, specification-dependent alignments exist on this fixed panel, that they are concentrated in embedding-geometry readouts rather than attention readouts, and that under this audit they cannot be distinguished from residual confound structure. A further observation points the same way: our own protocol pre-registers a readout sanity check in which a usable readout should correlate with co-expression (attention $\approx$ co-expression is well established). The supported positives come from readouts that fail that check (FM–co-expression Spearman -0.02 to -0.18), while the readouts that pass it (scFoundation $+0.08$, PBM Geneformer attention $+0.40$) are null. Embedding-cosine anisotropy interacting with degree conditioning is a more parsimonious account of that inversion than regulatory encoding. This is compatible with reports that scFM-derived gene relationships often resemble co-expression [7], and it does not license either a global regulatory-capability claim or a claim that FMs contain no regulatory information: the probe is linear and operates on pairwise graph summaries, and the proxy is itself a co-expression-family object.

Limitations and scope. The reference graph is a regulatory-potential proxy built from motif and accessibility evidence; proximity linkage, motif false positives, and cell-state aggregation remain limitations, and the proxy is near cell-type-invariant on this panel. The full specification conditions on proxy-derived degree covariates; the non-degree co-primary specification avoids that hazard and supports no row under both nulls. The panel is fixed rather than sampled, so p_{MC} describes unusualness under the stated randomizations, not uncertainty over a TF or tissue population, and no effect size carries an interval estimate. Model coverage is uneven across tissues and

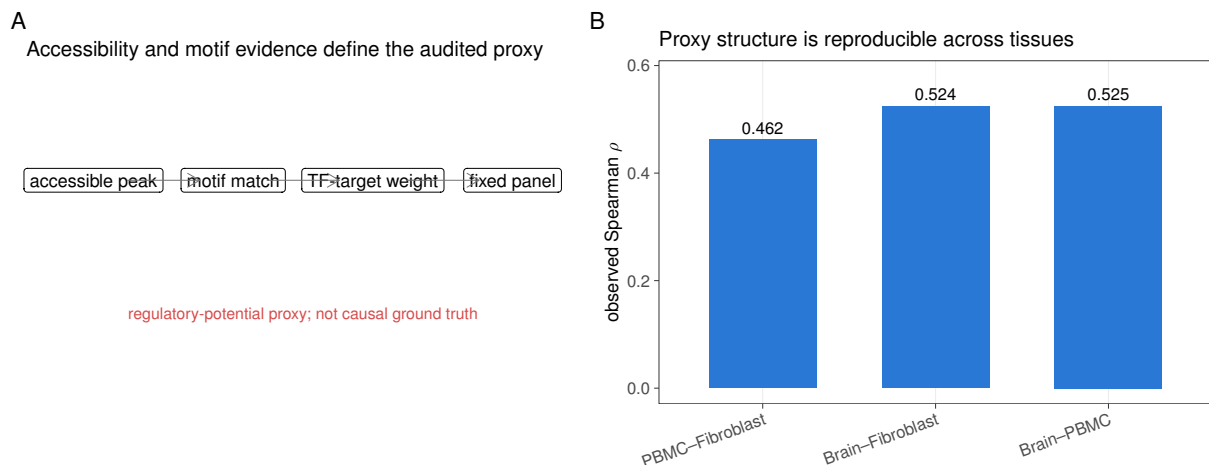


Figure 1: Regulatory-potential proxy and observed reproducibility. (A) The proxy combines accessible peaks, motif matches, and TF–target linkage on a fixed panel; it is not causal ground truth. (B) Observed consensus-graph correlations across three tissue datasets.

Tissue pair	Observed Spearman ρ	Shared TFs
Brain–PBMC	0.5249	446
Brain–Fibroblast	0.5245	446
PBMC–Fibroblast	0.4623	446

Table 1: Observed cross-tissue reproducibility of the regulatory-potential proxy. No confidence interval or new randomization inference is attached to these descriptive comparisons.

readouts. Per-cell-type and cross-tissue analyses are descriptive. The injection diagnostic is axis-aligned and cannot establish power against arbitrary biological alternatives. Fine-tuning, few-shot adaptation, decoder outputs, non-linear probes on raw per-gene embeddings, other gene panels, and direct perturbational validation are not tested.

5 Conclusion

scReg-Eval provides a reproducible fixed-panel protocol for separating co-expression and structural covariates from alignment with an accessibility/motif regulatory-potential proxy. Applied to thirteen pooled model/readout rows, it finds six small positive and one negative alignment supported by two randomization nulls, a co-expression baseline that reaches the same order of alignment, sign-inconsistent readouts within the same model, and positives that depend on conditioning on outcome-derived covariates. A supervised disjoint-TF probe gives mixed family-level contrasts, with the UCE and random-init floors distinguishing from co-expression but no model-specific regulatory recovery. The result motivates readout-specific, null-explicit, baseline-reported accounting rather than broad claims that scFMs either do or do not encode gene regulation.

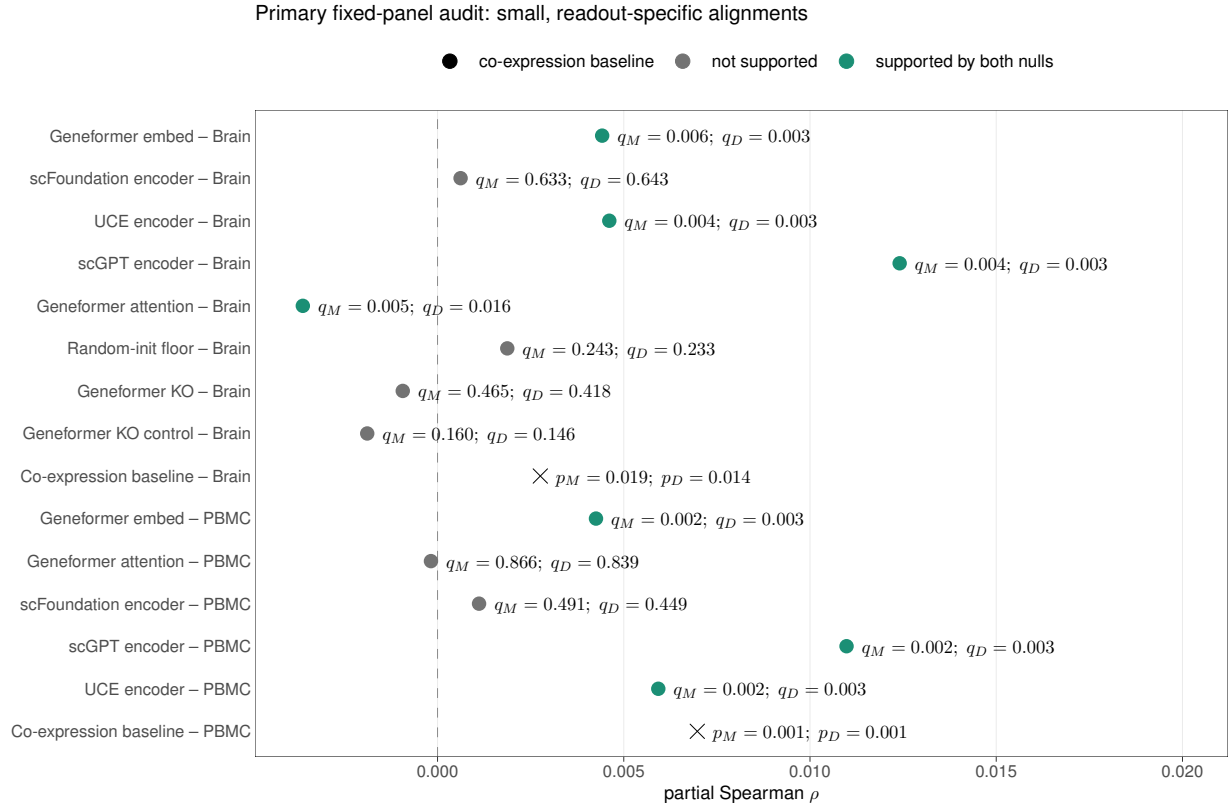


Figure 2: Primary full-confound fixed-panel audit. Points are observed partial Spearman correlations; labels give BH-adjusted gene-label (q_M) and row-shuffle (q_D) values within prespecified families. Green rows are supported by both nulls at $q < 0.05$. Crosses mark the co-expression baseline, tested outside the FM families with uncorrected p -values.

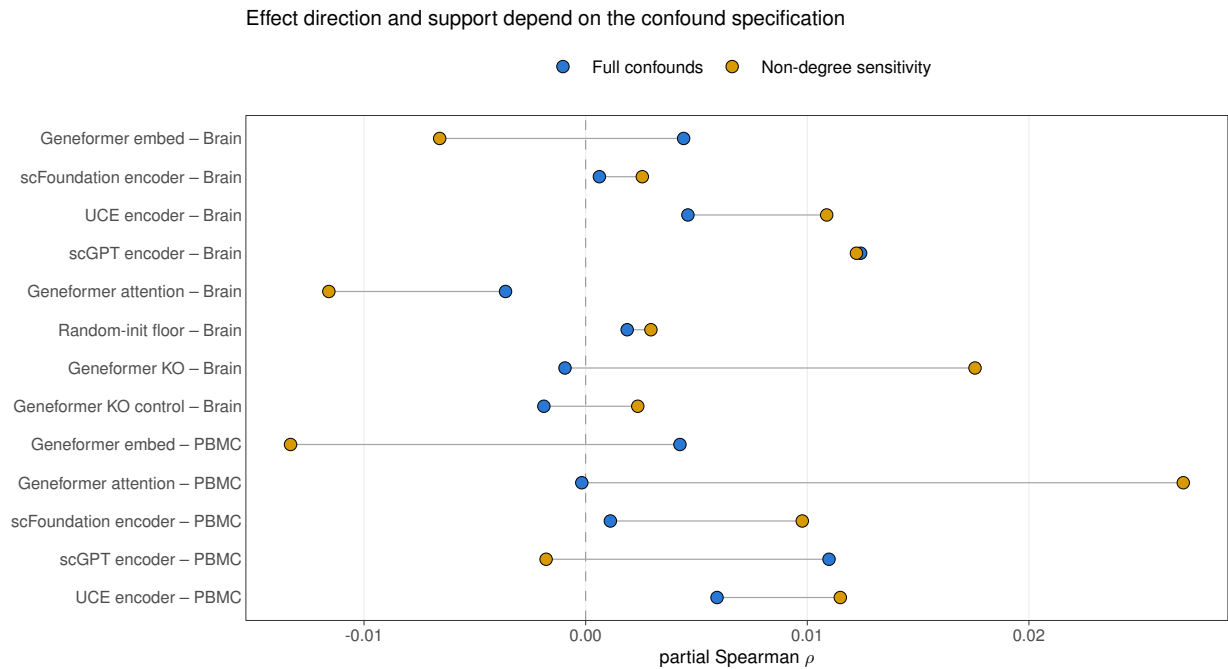


Figure 3: Confound-specification sensitivity. Full and non-degree estimates are joined within each model/readout and tissue. The non-degree results are secondary and show null-dependent support.

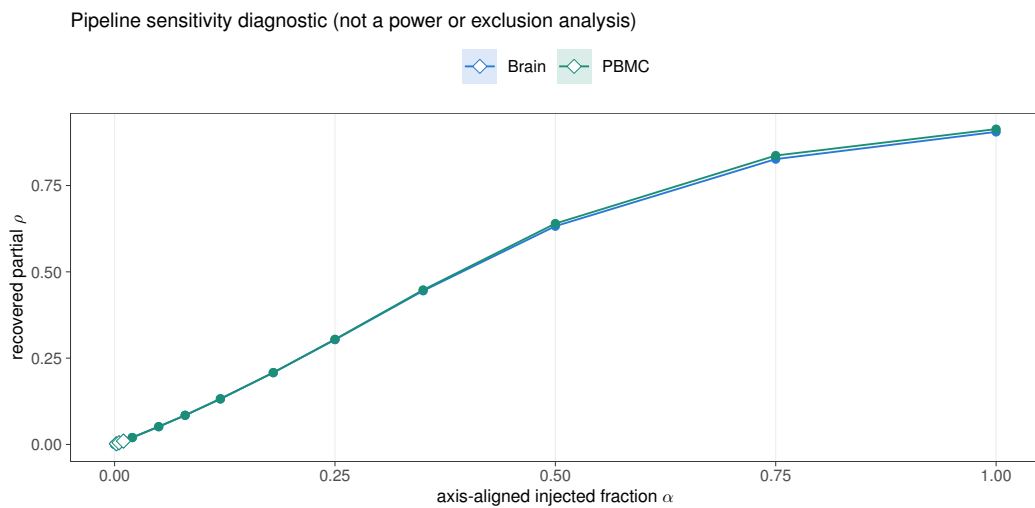


Figure 4: Axis-aligned pipeline-sensitivity diagnostic. Lines show the mean across 30 independent noise replicates and ribbons show the replicate range. This is not a confidence interval, power curve, MDE, or exclusion analysis.

Per-cell-type estimates are descriptive robustness checks

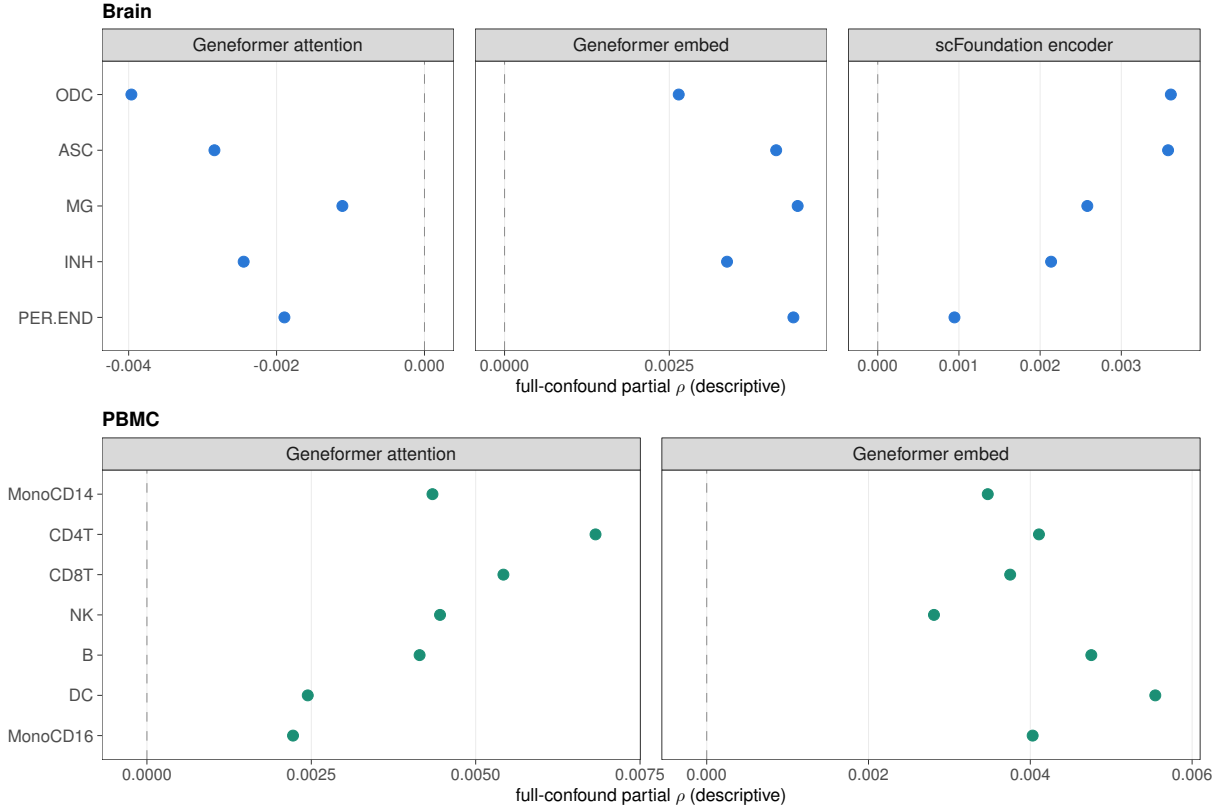


Figure 5: Per-cell-type full-confound estimates. These observed effects are descriptive exploratory robustness checks without randomization p -values or multiplicity claims.

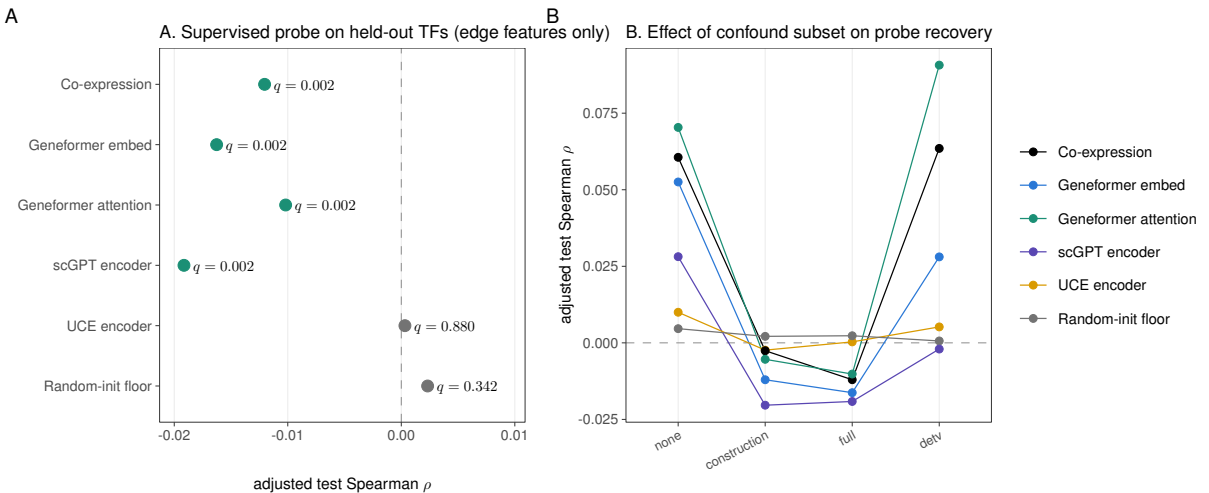


Figure 6: TF-disjoint supervised probe. (A) Confound-adjusted test correlation per family on 135 held-out TFs, with gene-label permutation q -values. Paired sign-flip tests against the co-expression baseline are supported for UCE ($q = 0.033$) and the random-init floor ($q = 0.030$) but not for Geneformer embedding, attention, or scGPT; the supported floor contrast prevents a model-specific reading of UCE. (B) The same readout across confound subsets; recovery collapses once the proxy's construction covariates (peak count, gene length, GC) are conditioned on, and random-init stays at zero throughout.

Tissue	Readout	Partial ρ	p_M	q_M	p_D	q_D
Brain	Geneformer embed	0.0044	0.003	0.006	0.001	0.003
Brain	scFoundation encoder	0.0006	0.633	0.633	0.643	0.643
Brain	UCE encoder	0.0046	0.001	0.004	0.001	0.003
Brain	scGPT encoder	0.0124	0.001	0.004	0.001	0.003
Brain	Geneformer attention	-0.0036	0.002	0.005	0.008	0.016
Brain	Random-init floor	0.0019	0.182	0.243	0.175	0.233
Brain	Geneformer KO	-0.0009	0.407	0.465	0.366	0.418
Brain	Geneformer KO control	-0.0019	0.100	0.160	0.091	0.146
Brain	Co-expression baseline	0.0028	0.019	—	0.014	—
PBMC	Geneformer embed	0.0043	0.001	0.002	0.002	0.003
PBMC	Geneformer attention	-0.0002	0.866	0.866	0.839	0.839
PBMC	scFoundation encoder	0.0011	0.393	0.491	0.359	0.449
PBMC	scGPT encoder	0.0110	0.001	0.002	0.001	0.003
PBMC	UCE encoder	0.0059	0.001	0.002	0.001	0.003
PBMC	Co-expression baseline	0.0070	0.001	—	0.001	—

Table 2: Primary fixed-panel results. M denotes the gene-label null and D the within-TF-row degree-preserving target shuffle; each q is BH-adjusted within its tissue-by-specification-by-null family. The co-expression baseline is tested on the same edge set outside the FM families; its entries are uncorrected p -values (dashes mark where no family correction applies).

Data and code availability

The fixed manifest, graph-construction and statistical code, deterministic seed metadata, complete model/readout coverage table, authoritative JSON outputs, and figure generator are provided in the project repository at <https://github.com/PeterPonyu/scfm-reg-audit> and archived at <https://doi.org/10.5281/zenodo.21724337>. Public inputs include GSE174367 brain data [10], the 10x Genomics 10k PBMC multiome [1], and GSE206767 fibroblast ATAC data. Large upstream datasets and model weights are not redistributed; the availability record documents their stable source identifiers and the exact manifests used.

Ethics statement

This study reuses public, de-identified datasets and performs no new human-subject recruitment or intervention.

Competing interests

The author declares no competing interests.

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CRedit author contributions: Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing, and Project administration.

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